

Project Design Phase-II Technology Stack (Architecture & Stack)

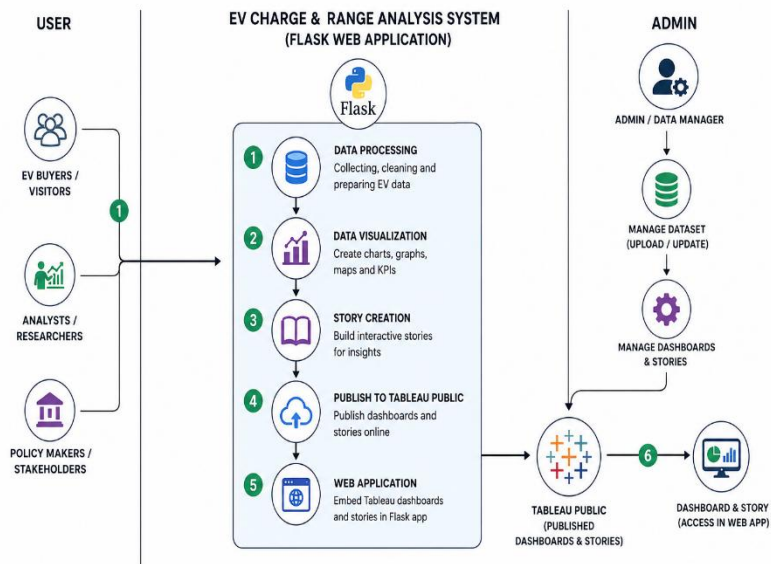
Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	EV dataset containing vehicle specifications, charging, range, price, and efficiency details.	CSV Dataset
2.	Data Preparation	Cleaning, formatting, and preprocessing the EV dataset before analysis.	Tableau Desktop
3.	Calculated Fields	Creation of calculated fields and KPIs for EV analysis.	Tableau Calculated Fields
4.	Data Visualization	Development of charts, graphs, maps, and analytical visualizations.	Tableau Desktop
5.	Dashboard Development	Creation of an interactive dashboard with KPIs, charts, and filters.	Tableau Dashboard
6.	Story Development	Presentation of EV insights using Tableau Story.	Tableau Story
7.	Dashboard Publishing	Publishing dashboards and stories for online access	Tableau Public
8.	User Interface	Responsive interface to access dashboards, stories, and project information.	HTML5, CSS3, Bootstrap, JavaScript

9.	Application Logic	Handles webpage navigation and embeds Tableau Dashboard and Story.	Python, Flask
10.	File Storage	Stores datasets, images, and project assets locally.	Local File System
11.	Infrastructure (Server)	Local deployment and execution of the web application	Flask Development Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source tools used for developing the application.	Flask, Bootstrap, Tableau Public
2.	Security Implementations	Secure routing and safe embedding of Tableau dashboards.	HTML5 Validation, Flask Routing
3.	Scalable Architecture	Supports additional datasets, dashboards, and future enhancements.	Three-Tier Architecture (Data Layer, Visualization Layer, Presentation Layer)
4.	Availability	Dashboard and Story are published online and accessible anytime.	Tableau Public, Flask
5.	Performance	Optimized dashboards with responsive design and fast loading.	Tableau Public, Bootstrap, JavaScript

References:

- <https://www.tableau.com/>
- <https://help.tableau.com/>
- <https://flask.palletsprojects.com/>
- <https://developer.mozilla.org/>